Project Design Phase-II
Data Flow Diagram & User Stories

Date	27 February 2026
Team ID	699d5f89c40b5ad0e9ee5981
Project Name	QUOTES RECOMMENDATION CHATBOT USING NLP
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a graphical representation that shows how data moves through a system. It illustrates how data enters the system, how it is processed, and where it is stored.

For the Quotes Recommendation Chatbot, the DFD represents the interaction between the user, the NLP processing module, the quotes database, and the chatbot response system.

DFD Level 0 (Context Level)

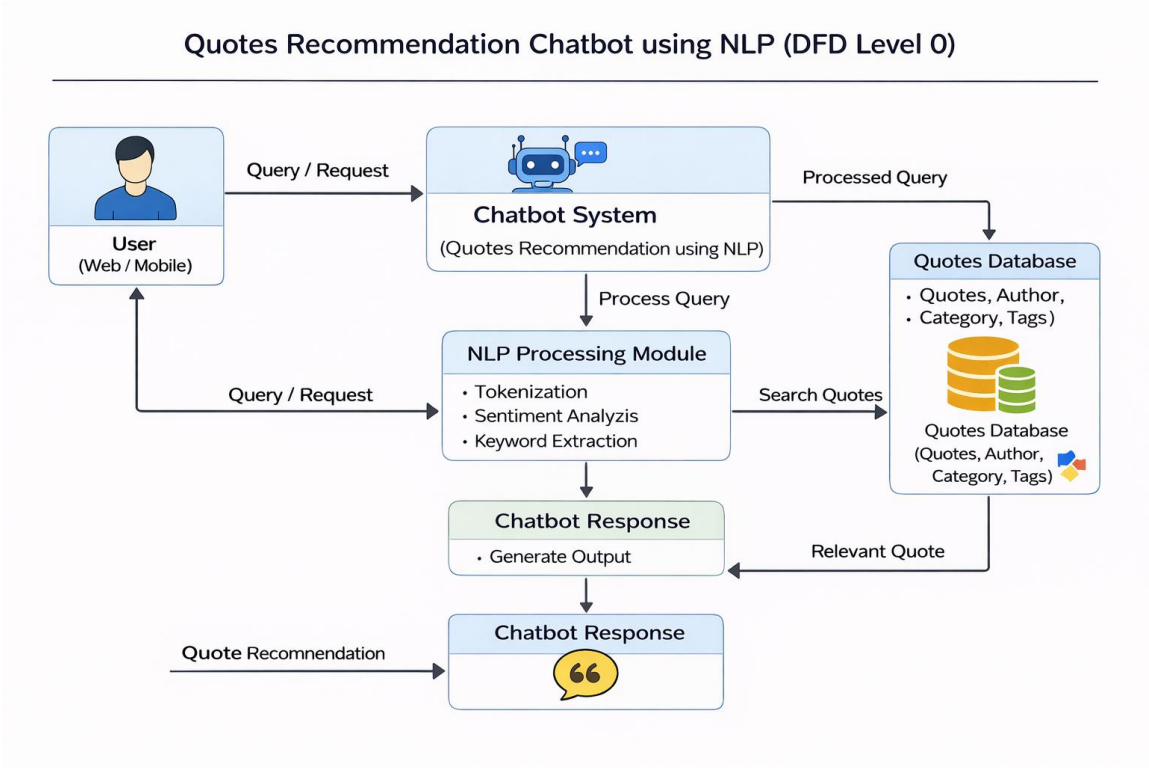
Main Components:

1. **User**
   The user enters a text query or request for a quote.
2. **Chatbot System**
   The system receives and processes the user input using NLP techniques.
3. **NLP Processing Module**
   The module performs text preprocessing, tokenization, sentiment analysis, and keyword extraction.
4. **Quotes Database**
   The system retrieves relevant quotes based on the processed input.

5. **Chatbot Response**
   The recommended quote is returned to the user as a response.

Data Flow:

User → Input Query → NLP Processing → Quote Retrieval → Response Generation → User



The DFD Level 0 represents the overall data flow of the Quotes Recommendation Chatbot system.

1. The User enters a query or request for a motivational quote.
2. The Chatbot System receives the input.

3. The query is processed by the NLP Processing Module, which performs text preprocessing, sentiment analysis, and keyword extraction.
4. The system searches the Quotes Database to find relevant quotes based on the analyzed input.
5. The Chatbot Response Module generates the final response.
6. The recommended quote is returned to the User.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Web User)	Chatbot Interaction	USN-1	As a user, I want to enter a message so that the chatbot can recommend a relevant quote.	User can submit query and receive quote response	High	Sprint-1
Customer (Web User)	NLP Processing	USN-2	As a user, I want the system to understand my emotions or keywords so that the chatbot provides meaningful quotes.	System successfully analyzes sentiment and keywords	High	Sprint-1
Customer (Web User)	Quote Recommendation	USN-3	As a user, I want the chatbot to recommend motivational or inspirational quotes based on my input.	System retrieves appropriate quote from database	High	Sprint-2
Customer (Web User)	Response Display	USN-4	As a user, I want the chatbot to display the recommended quote clearly so that I can read it easily.	Quote and author are displayed properly	Medium	Sprint-2
Customer (Web User)	Chatbot Interface	USN-5	As a user, I want a simple chatbot interface so that interaction is easy and user-friendly.	User interface allows easy interaction	High	Sprint-1